

Program: Diploma in Civil Engineering and Planning	
Course Code: 6398	Course Title: Basic Geotechnical Engineering Lab
Semester: 4	Credits: Nil
Course Category: Program Core	
Periods per week: 3 (L:0, T:1 P:3)	Periods per semester: 60

Course Objectives:

- To provide hands-on experience with the laboratory tests used for determination of physical, index and Engineering properties of soil.
- To provide expertise to handle different Soil testing equipment and conduct field tests

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of basic Mathematics		Engineering Mathematics	2
Basics of soil mechanics		Basic Geotechnical Engineering	4

Course Outcomes:

On completion of the course, the student will be able to:

CO	Description	Duration (Hours)	Cognitive level
CO1	Determine the index properties of soil.	12	Understanding
CO2	Classify soil based on index properties as per Indian standards.	15	Understanding
CO3	Determine the coefficient of permeability, consolidation characteristics of soil and illustrate field methods of determination of bearing capacity	14	Applying

CO4	Determine shear strength parameters and compaction characteristics of soil.	15	Applying
	Lab test	4	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3			
CO2				3			
CO3				3			
CO4				3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Determine the index properties of soil		
M1.01	Determine water content of given soil sample by oven drying method as per IS: 2720 (Part-II).	1	Understanding
M1.02	Determine specific gravity of soil by pycnometer method as per IS 2720 (Part- III).	1	Understanding
M1.03	Determine specific gravity of soil by density bottle method as per IS 2720 (Part- III).	2	Understanding
M1.04	Determine dry unit weight of soil in field by core cutter method as per IS 2720 (Part-XXIX).	3	Understanding
M1.05	Determine dry unit weight of soil in field by sand replacement method as per IS 2720 (Part-XXVIII).	3	Understanding
M1.06	Prepare a detailed report on the method of preparation of dry samples from the bulk soil sample received from the field for various laboratory tests.	2	Understanding
CO2	Classify soil based on index properties as per Indian standards.		

M2.01	Determine Plastic and Liquid Limit along with Plasticity Index of given soil sample as per IS 2720 – 1985 (Part- V).	3	Understanding
M2.02	Determine Shrinkage limit of given soil sample as per IS 2720 (Part- VI).	3	Understanding
M2.03	Determine grain size distribution of given soil sample by wet and dry sieve analysis as per IS 2720 (Part- IV).	3	Understanding
M2.04	Determine grain size distribution of given fine grained soil sample by Hydrometer analysis as per IS 3104 -1982	3	Understanding
M2.05	Classify soil by conducting field tests through Visual inspection.	3	Understanding
	Lab Test	2	
CO3	Determine the coefficient of permeability, consolidation characteristics of soil and illustrate field methods of determination of bearing capacity		
M3.01	Determine coefficient of permeability by constant head test as per IS 2720 (Part- XVII).	3	Understanding
M3.02	Determine coefficient of permeability by falling head test as per IS 2720 (Part- XVII).	3	Understanding
M3.03	Determination of coefficient of consolidation by conducting consolidation test as per IS 2720- 1986 (Part XV)	3	Understanding
M3.04	Prepare a detailed report on plate load test after conducting a site visit	3	Applying
M3.05	Prepare a detailed report on Standard Penetration Test after conducting a site visit	2	Applying
CO4	Determine shear strength parameters and compaction characteristics of soil.		
M4.01	Determine shear strength of soil by direct shear test as per IS 2720 (Part-XIII).	3	Understanding
M4.02	Determine unconfined compressive strength of soil as per IS 2720-part X	3	Understanding
M4.03	Determine MDD and OMC by standard proctor test of given soil sample as per IS 2720 (Part- VII).	3	Understanding
M4.04	Determination of CBR value on the field as per IS2720 (Part - XVI).	3	Understanding
M4.05	Open ended experiment	3	Applying
	Lab Test	2	

Text / Reference

T/R	Book Title/Author
T1	Punmia, B.C., Soil Mechanics and Foundation Engineering, Laxmi Publication, Delhi.
T2	Gopal Ranjan &A.S.R.Rao, Basic and applied soil mechanics, New Age International
T3	Arora K R, Soil Mechanics and Foundation Engineering, Standard Publisher.
R1	Murthy, V.N.S., A text book of soil mechanics and foundation Engineering, CBS Publishers & Distributors Pvt. Ltd., New Delhi.
R2	Ramamurthy, T.N. &Sitharam,T.G., Geotechnical Engineering(Soil Mechanics), S Chand and Company LTD., New Delhi.
R3	Raj, P. Purushothama, Soil Mechanics and Foundation Engineering, Pearson India, New Delhi.
R4	Kasamalkar, B. J., Geotechnical Engineering, Pune Vidyarthi GrihaPrakashan, Pune.

Online Resources:

Sl No	Website Link
1	https://www.nptel.ac.in/courses/105106142
2	https://nptel.ac.in/courses/105101160
3	https://www.nptel.ac.in/courses/105101001
4	https://www.astm.org/Standards/geotechnical-engineering-standards.html